

PFC AI Design Agent Documentation Improvement Plan

A comprehensive roadmap for improving the technical report (Steps 1–16),
in-code documentation, and developer guides

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Prepared for: Engineering Team — PFC Power Stage Design Tool

This document captures the full scope of documentation improvements agreed during the planning discussion of June 2026. It serves as the master reference before any code changes are made. It describes **what** needs to be added to each section of the generated report and developer guides, **why** it is needed, and **who** it serves.

No source files are modified by this document. It is a planning and alignment artifact only.

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1. Governing Principle — Progressive Disclosure

The central challenge in improving this documentation is that the intended readership spans an enormous range of background knowledge — from a first-year electrical engineering student encountering PFC for the first time, to a principal power electronics engineer who designs converters daily, to an academic reviewer validating the calculation methodology. A single document must serve all of these readers without either overwhelming the novice or boring the expert.

The solution is **Progressive Disclosure**: every section of every step is structured in four distinct layers. Readers consume as many layers as their background requires and skip the rest. The layers are never mixed — each one is self-contained.

The Four-Layer Structure

Layer	Name	Content	Length	Primary Reader
1	Plain-English Objective	What does this step achieve? Why does it matter to the final design?	2–4 sentences	All audiences — sets context
2	Governing Equations	Every variable defined with name, symbol, unit, and typical range. Source reference cited.	½–1 page	Junior & mid-level engineer
3	Worked Calculation	Actual design numbers substituted step by step. Intermediate results shown. Standard-value rounding explained.	¼–½ page	All engineering levels
4	Result Interpretation	Pass/Fail verdict. What the number means physically. Sensitivity — what happens if this value is 20% higher or lower?	2–5 sentences	Senior engineer, professor

Why not write two separate documents — one introductory, one advanced?

Separate documents diverge over time. When the calculation engine is updated, only one version gets updated. The layered single-document approach keeps everything synchronised and allows the tool to produce one authoritative report that serves all stakeholders simultaneously — a review meeting can contain both a student intern and a chief engineer around the same table, reading the same document.

2. Target Audience Analysis

Before specifying what content to add, it is useful to define precisely what each audience segment needs from the report and where the current report falls short for them.

Audience	Background Assumed	Primary Need from Report	Current Gap
College student / intern	Circuit theory, Ohm's Law, basic magnetics. Little or no switching converter experience.	Understand what each block does physically. Follow the chain of logic from spec to component value.	Equations appear without derivation. Terms like "duty cycle" and "RHP zero" are used without definition.
Junior engineer (0–3 yrs)	Knows switching converters at system level. Can simulate but has not designed magnetics or control loops from scratch.	See the full design methodology. Understand why each design choice was made, not just what was chosen.	Calculation results shown but reasoning absent. Cannot reproduce or adapt the calculation independently.
Mid-level engineer (3–8 yrs)	Has designed PFC stages. Comfortable with Bode plots, magnetics, thermal analysis.	Verify design margins and assumptions. Quickly check that correct formulas and standards were applied.	Some references are missing (IEC, SEMI, app notes). Assumptions not always stated explicitly.
Principal / chief engineer	Deep domain expert. Has reviewed hundreds of PFC designs.	Spot check critical decisions. Confirm all design rules and safety margins are met. Approve for production.	The design audit (Section 13.8) exists but does not trace each criterion back to a standard or a physical limit.
PhD professor / researcher	Expert in power electronics theory. Familiar with all models cited.	Evaluate model accuracy. Assess whether iGSE, SA thermal model, and SLVA662 Method B are applied correctly.	Model selection rationale and limitations are not documented. No statement of model uncertainty.

3. PDF Report — Steps 1–6 (System-Level Design)

Steps 1 through 6 establish the design requirements and select the power stage topology. They are the foundation on which every subsequent calculation depends. Currently, these steps present results clearly but provide little context about where the requirements come from or why the chosen topology and operating parameters are appropriate.

Step 1 — Design Requirements & Standards

What needs to be added

- **Why PFC is required:** A one-paragraph explanation of harmonic distortion in uncontrolled rectifiers, the role of IEC 61000-3-2 in limiting harmonic content, and what Power Factor physically means (ratio of real to apparent power, phasor diagram).
- **SEMI F47 explanation:** Define voltage sag ride-through requirements — what a "sag" is, the 200 ms / 70% Vnom test condition, and how the hold-up capacitor directly determines compliance.
- **Operating conditions table narrative:** Explain why the 9-point operating matrix (4 low-line + 5 high-line) is the standard characterisation grid for universal-input PFC designs, and which corner drives each design element (90 Vac for magnetics worst-case, 180 Vac for capacitor ripple current worst-case).
- **Specification traceability table:** Link every input requirement (output voltage, output power, hold-up time, efficiency) to the customer requirement or standard that mandates it.

Steps 2–3 — Topology & Mode Selection

What needs to be added

- **Boost converter operating principle:** For the student reader — explain why a boost topology is universal for PFC (input voltage always below output, continuous input current available from inductor).
- **Interleaved vs. single-phase comparison table:** Quantify the tradeoff — ripple current reduction factor ($\times 4$ at 50% duty for 2-phase), inductor size reduction, gate drive complexity increase, EMI filter size reduction.
- **CCM vs. CrCM boundary:** Define both modes. Explain why CCM is preferred at higher power levels (lower peak current, lower RMS current in capacitor) and why CrCM is preferred at lower power (zero-current switching, no reverse recovery).

Steps 4–5 — Operating Point & Duty Cycle Calculation

What needs to be added

- **Duty cycle derivation:** Show the full derivation $D = 1 - V_{pk}/V_{out}$ from volt-second balance on the inductor. Explain why D varies with line voltage and where in the line cycle duty cycle is maximum (at line

voltage zero crossing).

- **Peak current derivation:** Show the $I_{pk} = \sqrt{2} \times P_{out} / (\eta \times PF \times V_{in_rms})$ formula and explain each term, particularly the power factor term and why it appears.
- **Worked example for each operating corner:** A compact table showing D , V_{in_pk} , I_{pk} for all 9 operating points with a colour highlight on the worst-case values.

Step 6 — Switching Frequency Selection

What needs to be added

- **The four-way frequency tradeoff:** Core loss $\propto f^\alpha$, copper loss $\propto f^{0.5}$ (skin effect), gate drive loss $\propto f$, EMI filter size $\propto 1/f$. Show these competing trends in a qualitative plot or table and explain where 70 kHz sits on each axis.
- **Resonant frequency constraints:** Explain why $f_{sw}/f_{ci} \geq 6$ and why the LC pole of the output filter must be well below the voltage-loop crossover.

4. PDF Report — Steps 7–8 (Magnetic Design)

The magnetic design steps are the technical heart of the PFC design. They currently contain the most calculation depth but the least explanatory narrative. A junior engineer reading these sections cannot understand why a powder core is chosen over ferrite, what the fill factor limit means physically, or why the iGSE correction is needed. This section requires the most significant documentation investment.

Priority: HIGHEST

Steps 7–8 are where the tool adds the most unique value — the magnetic sizing engine is a full custom implementation. The documentation here determines whether a reviewer trusts the output or questions every number. Invest heavily here.

Step 7 — Background Narrative (new — approximately 1 page)

This narrative should appear once at the start of Step 7 and set the conceptual framework for everything that follows. It does not exist today.

Sub-topics to cover:

- **Role of the PFC inductor:** It stores energy during the switch-on phase and transfers it to the output during switch-off. The inductance value determines the ripple current, which determines every loss term downstream.
- **Why toroidal powder cores:** Powder cores (Kool M μ , EDGE, XFlux) have a distributed air gap built into the material — permeability drops gradually with applied DC field (soft saturation). This is in contrast to ferrite cores which have a discrete air gap and a hard saturation cliff. For a PFC inductor carrying large DC bias, soft saturation means the design is more tolerant of current peaks and the inductance rolloff is predictable.
- **Core geometry — toroid vs. E-core:** Toroids have lower leakage flux and better EMI performance. E-cores are easier to wind but require a discrete gap. The tool uses toroidal powder cores as primary candidates.
- **The window fill factor — physical meaning:** The core window (centre hole of a toroid) has a cross-sectional area W_a . The conductor bundle occupies some fraction of this area. $FF_{cu} = N \times A_{conductor} / W_a$. Practical limits: 40% for easy hand winding, up to 65–70% with machine winding. Beyond 70%, the winding is physically impossible to fit. The fill factor is the single most common reason a design fails.
- **Stacking toroids:** When a single core cannot meet the inductance requirement with acceptable fill factor, two or three cores are stacked on the same winding. Critical: the window area does NOT multiply — each core retains its own bore. This is a frequently misunderstood point. The AL value (inductance per turn squared) multiplies with stacks because total magnetic length increases.

Step 7 — Core Loss Model (iGSE)

What needs to be added

- **The Steinmetz Equation — origin and limitations:** The original Steinmetz equation $P_{core} = K \times B_{pk}^{\alpha} \times f^{\beta}$ applies to sinusoidal flux waveforms. A PFC inductor carries trapezoidal flux — the waveform has two slopes: a slow charge phase (switch on) and a fast discharge phase (switch off). The Steinmetz equation applied at peak B overestimates loss at high line (D small, fast transitions dominate) and underestimates at low line. The error can be 30–60%.
- **The iGSE correction F(D):** The Improved Generalised Steinmetz Equation applies a duty-cycle correction factor $F(D) = K \times [D^{\alpha}(1-c) + (1-D)^{\alpha}(1-c)]$. At 50% duty $F(D) \approx 1$, at 5% duty $F(D) \approx 1.6$, at 95% duty $F(D) \approx 1.6$. The correction is largest at the extremes of duty cycle — exactly where the PFC operates at high and low line voltages.
- **Material-specific constants:** The Steinmetz constants α , β , K are fitted to manufacturer core loss data at the operating frequency and temperature. The tool uses a proprietary database of fitted constants. The derivation of $a_{effective}$ from measured loss data is shown in the Review section.

Step 7 — Thermal Model

What needs to be added

- **Why the SA power-law model rather than Rth:** The thermal resistance model ($\Delta T = P \times R_{th}$) assumes a fixed linear relationship between power and temperature rise. For natural-convection cooling of a toroidal inductor, the heat transfer coefficient itself varies with temperature difference — the relationship is non-linear. The surface-area power-law $\Delta T = (P \times 1000 / SA_{cm^2})^{0.833}$ captures this non-linearity and has been validated against measured inductor temperatures. The 0.833 exponent comes from natural convection scaling laws (Churchill-Chu correlation).
- **Surface area calculation:** For a wound toroid the radiating surface is the outer cylindrical surface ($OD \times height$), the two annular end faces, and the inner bore cylindrical surface reduced by the winding. The formula is given with all geometric terms defined.

Step 8 — Time-Domain Core Loss

What needs to be added

- **Why time-domain integration rather than peak-point calculation:** The crest of the half line cycle (peak input current) gives maximum B and maximum P_{core} at that instant. But this is not where the inductor operates most of the time. The thermal performance is governed by the average loss over the full half cycle. The ratio $P_{core_avg} / P_{core_crest}$ varies from 0.5 to 0.8 depending on the operating corner — using crest-only calculation can overestimate thermal dissipation by 25–50%.
- **Half-cycle waveform construction:** Explain the 360-point waveform — at each point in the half line cycle, the instantaneous input current (sine envelope) gives the DC operating point, from which H_{dc} , B_{dc} , and $B_{ac,pk}$ are computed, from which instantaneous P_{core} is computed via iGSE.

- **Avg/Crest ratio table interpretation:** The table showing this ratio for each operating point is already in the report — add a column explaining what a ratio < 0.7 means (high line dominant, crest-only would significantly overestimate) vs. > 0.9 (crest is representative of the whole cycle).

5. PDF Report — Steps 9–12 (Intermediate Calculations)

Steps 9 through 12 cover wire selection, copper loss, thermal analysis, and efficiency roll-up. These sections currently present results with minimal narrative — they are the most underdeveloped part of the report. Each sub-step needs both a conceptual introduction and a result interpretation paragraph.

Step 9 — Wire Selection

What needs to be added

- **Skin effect and proximity effect — definitions:** At switching frequency, current crowds to the conductor surface (skin depth $\delta = \sqrt{\rho / \pi f \mu}$ ≈ 0.25 mm at 70 kHz for copper). A solid conductor larger than 2δ in diameter wastes its cross section. Proximity effect: adjacent conductors carrying opposing currents induce additional eddy currents, multiplying the effective AC resistance by a factor R_{ac}/R_{dc} .
- **Why Litz wire:** Litz wire subdivides the conductor into many fine strands (each smaller than δ), twisted so that each strand experiences the same average flux linkage. The goal is to keep each strand diameter below 2δ at the switching frequency. The strand count formula is derived from the required copper area divided by the area of one strand.
- **Current density target:** $J_{\text{target}} = 3\text{--}5$ A/mm² for natural convection cooling. The tool uses 3.5 A/mm² as default. Explain why higher J means smaller wire (better fill factor) but higher loss and temperature.

Step 10 — Copper Loss Analysis

What needs to be added

- **DC vs. AC copper loss decomposition:** $P_{cu} = P_{cu_dc} + P_{cu_ac} = I_{dc}^2 \times R_{dc} + I_{hf_rms}^2 \times R_{ac}$. Show that for a PFC inductor at 70 kHz with Litz wire, P_{cu_ac} is typically 5–15% of P_{cu_dc} — significant but not dominant. Without Litz it would be 30–60%.
- **Temperature coefficient of copper:** $R_{dc} \text{ at } 100^\circ\text{C} = R_{dc_20^\circ\text{C}} \times (1 + 0.00393 \times 80) = 1.31 \times R_{dc_20^\circ\text{C}}$. Explain why the 100°C evaluation temperature is chosen — it is the thermal design point, not an absolute maximum.

Steps 11–12 — Efficiency and Loss Roll-Up

What needs to be added

- **Loss budget table with percentage breakdown:** A pie-chart equivalent in table form — core loss, copper loss, thermal headroom — with percentage of total budget for each loss term. This allows a reader to immediately see the dominant loss mechanism.
- **Temperature rise budget explained:** Explain the $\Delta T_{\text{budget}} = 40^\circ\text{C}$ (for a 60°C ambient, limiting core to 100°C). Show the margin: $\Delta T_{\text{design}} / \Delta T_{\text{budget}}$.

6. PDF Report — Steps 13–14 (Detailed Magnetic Analysis)

Steps 13 and 14 are currently the most technically detailed sections of the report. The calculations are correct and comprehensive. The improvement needed here is not more calculation but better narrative framing — the reader needs to understand what the tables are telling them and how to use the results to make engineering decisions.

Step 13 — Candidate Core Comparison

What needs to be added

- **Selection criterion narrative before the table:** A paragraph explaining which metrics were used to rank candidates (AL tolerance, Ku fill, ΔT , loss, geometry) and what weighting philosophy was applied. Why is "Best overall" defined as it is?
- **Per-candidate engineering commentary:** After the candidate table, add one sentence per candidate explaining what distinguishes it. Example: "The 3-stack design achieves lowest total loss but has the largest footprint — preferred for open-chassis designs." "The 2-stack design is the most compact but operates 8°C closer to the thermal limit."
- **Approved design highlight with rationale:** A separate callout box (already highlighted in blue) should include one paragraph: "This design was selected because ... It provides X% margin on temperature, Y% margin on saturation, and Z% margin on fill factor."

Step 13 — Design Audit Checklist

What needs to be added

- **Source citation for each criterion:** Each PASS/FAIL row should cite the reason: "FFcu < 65% — IEC 60317 winding practice guideline", " $\Delta T < 40^{\circ}\text{C}$ — thermal budget for 100°C max core temp at 60°C ambient", " $B_{\text{max}} < 0.8 \times B_{\text{sat}}$ — 20% saturation margin per MAGNETICS application note."
- **Sensitivity note for FAIL conditions:** When a criterion fails, add a note explaining the simplest corrective action: "To reduce ΔT by 5°C, reduce N by 2 turns or increase core stack count to 3."

Step 14 — Extended Waveform Analysis

What needs to be added

- **How to read the waveform plots:** A paragraph before the first waveform explaining the x-axis (one half-cycle = 8.33 ms for 60 Hz), what H(t) represents (magnetising force as a function of instantaneous current), and why B(t) lags H(t) (the B-H loop of the material).
- **Loss uncertainty band interpretation:** "The ± 5 –20% band reflects three uncertainty sources: (1) Steinmetz constant fitting error from the manufacturer data (typically $\pm 5\%$), (2) temperature variation

within the operating range ($\pm 8\%$), (3) waveform shape approximation in the half-cycle model ($\pm 7\%$). The design is considered acceptable when the upper bound of the band is below the thermal limit."

7. PDF Report — Step 15 (Capacitor Design)

Step 15 sizes the DC bus output capacitor for hold-up, ripple voltage, and ripple current requirements, then selects a real part from the database and estimates service life. The calculations are present but the methodology and physical reasoning are not explained, leaving the reader unable to verify the approach independently.

Step 15.1–15.4 — Capacitor Sizing Theory

What needs to be added

- **Hold-up time energy balance derivation:** Show the full derivation from first principles: The capacitor must supply all load energy during the hold-up interval. $\frac{1}{2}C \times (V_{\text{max}}^2 - V_{\text{min}}^2) = P_{\text{out}} \times t_{\text{hold}} \rightarrow C_{\text{holdup}} = 2 \times P_{\text{out}} \times t_{\text{hold}} / (V_{\text{max}}^2 - V_{\text{min}}^2)$. Define every variable. Explain where V_{min} comes from (SEMI F47 specifies 70% of nominal for the 200 ms sag condition).
- **Ripple voltage sizing:** The second sizing criterion. Explain that the bus voltage ripple appears as a 120 Hz oscillation ($2 \times$ line frequency) and that excessive ripple degrades the quality of the output voltage regulation and increases harmonic content of the input current. $C_{\text{ripple}} = P_{\text{out}} / (2\pi \times f_{\text{line}} \times V_{\text{out}} \times \Delta V_{\text{pp}})$.
- **Governing capacitance — why the larger of the two:** The capacitor must satisfy both criteria simultaneously. The governing (larger) value determines the bank size.

Step 15.5–15.8 — Ripple Current & Part Verification

What needs to be added

- **LF + HF ripple current decomposition:** The capacitor carries two superimposed current components — a large-amplitude 120 Hz component from the load power pulsation, and a smaller high-frequency component at the switching frequency from the inductor current ripple. These add in RMS: $I_{\text{total}} = \sqrt{I_{\text{LF}}^2 + I_{\text{HF}}^2}$.
- **Why worst case is high-line, full load:** At high line and full load, the rectified input current has the largest 120 Hz ripple component (highest power, poorest power factor correction). The RMS current in the capacitor is maximum here.

Step 15.9–15.16 — Lifetime Analysis

What needs to be added

- **The Arrhenius lifetime model — physical basis:** Electrolytic capacitor aging is dominated by electrolyte evaporation through the rubber seal. This is a thermally activated chemical process following the Arrhenius equation. The simplified form: $\text{Life} \approx L_{\text{ref}} \times 2^{((T_{\text{max}} - T_{\text{c}})/10^\circ\text{C})}$. Every 10°C increase in core

temperature halves the expected life. This is the most cited capacitor lifetime rule of thumb.

- **Three-method comparison explained:** The tool computes life with three methods (Rubycon formula, Nippon Chemi-Con formula, and simplified Arrhenius) and reports the minimum. Explain why three methods: different manufacturers use different reference conditions in their datasheets. Taking the minimum is conservative and accounts for manufacturing variation.
- **15-year target — rationale:** Industrial equipment (data centres, UPS systems) typically carries a 15-year design life requirement. This is not a regulatory requirement but a commonly adopted industry design target for infrastructure power supplies.

8. PDF Report — Step 16 (Control Loop Design)

Step 16 is the most conceptually demanding section of the entire report. Control loop design requires understanding of feedback systems, Bode analysis, and AC small-signal modelling — topics that many competent power electronics engineers have limited exposure to beyond what they use day-to-day. This section requires the most substantial new content investment.

Priority: HIGH — largest documentation gap

The current Step 16 shows component values and Bode plots with almost no context. A reader who has not designed a control loop cannot interpret the plots or understand what Phase Margin means in practice. A reader who is expert in control theory cannot verify the methodology because the derivation is not shown. Both groups are underserved.

Step 16 — Background Section (new — approximately 1.5 pages)

This background section should appear once at the start of Step 16. It does not exist today.

16.0.1 — What is Average Current Mode (ACM) Control?

Explain the two-loop structure with plain language before any equations:

- **Inner current loop:** Forces the average inductor current to follow a reference signal that is shaped like a rectified sine. This makes the input impedance look resistive — the fundamental mechanism of power factor correction.
- **Outer voltage loop:** Adjusts the amplitude of the current reference to maintain the output voltage at the setpoint. It operates at a much lower bandwidth than the current loop (typically 10× lower frequency).
- **The GMOD block:** The gain modulator scales the current reference with the input voltage to maintain a constant power-bandwidth product across the line cycle. Without GMOD, the loop gain would vary 9:1 over the line cycle ($90V^2 : 264V^2$).

16.0.2 — What is a Bode Plot and How to Read It?

For readers unfamiliar with frequency-domain analysis:

- **Magnitude plot:** Shows how much the loop amplifies or attenuates signals at each frequency. 0 dB = unity gain. Positive dB = amplification. Negative dB = attenuation. The crossover frequency f_c is where the magnitude crosses 0 dB — this is the bandwidth of the loop.
- **Phase plot:** Shows the phase shift introduced by the loop at each frequency. A phase of -180° means the feedback has become positive — the system will oscillate if there is any gain at this frequency.
- **Phase Margin (PM):** At the crossover frequency, how far is the phase from -180° ? $PM = 180^\circ + \angle T(j\omega_c)$. A PM of 45° means good stability with moderate overshoot (~20%). PM of 60° means excellent stability with

minimal overshoot (~8%). $PM < 30^\circ$ is considered marginal; $PM < 0^\circ$ means the system is unstable.

- **Gain Margin (GM):** At the frequency where phase = -180° , how many dB of gain must the loop gain increase before instability? $GM > 6$ dB is the typical minimum.

16.0.3 — RHP Zeros: The Fundamental Bandwidth Limitation

This concept is often poorly understood. A clear explanation is essential:

- **Physical origin:** When the duty cycle increases momentarily (to deliver more energy), the inductor stores more energy but the output capacitor initially discharges slightly (less current flows to load at the instant of duty cycle increase, before the inductor current builds up). The output voltage initially moves in the wrong direction — this is the right-half-plane zero effect.
- **Why it limits bandwidth:** A RHP zero adds phase lag to the loop (like a pole) but adds magnitude gain (like a zero). If the loop bandwidth is set above f_{RHPz} , the phase lag causes instability. The safe rule is $f_{\text{cv}} < f_{\text{RHPz}} / 5$, providing 14 dB of gain margin from the RHP zero alone.
- **Location at different operating points:** $f_{\text{RHPz}} = RL \times Dp^2 / (2\pi \times L)$. At low line (Dp large), f_{RHPz} is low — most restrictive. At high line (Dp small), f_{RHPz} is large — less restrictive. The low-line corner always governs bandwidth selection.

Step 16.2 — Current Loop Compensator Design

What needs to be added

- **Type-II compensator topology explanation:** The Type-II OTA has one integrator (for DC accuracy — zero steady-state error), one zero (for phase boost at crossover), and one high-frequency pole (to attenuate switching noise). Show the circuit schematic topology: RIC in series with C1, then C2 across the RIC-C1 chain, all on the IEA output pin.
- **Phase boost calculation shown:** The boost provided = $\arctan(f_c/f_z) - \arctan(f_c/f_p)$. Show this evaluated at the design crossover frequency with the designed component values. Compare to the required boost ($90^\circ - (180^\circ + \angle Ti_{unc}(j\omega_{ci}))$).
- **Why Type-II is always sufficient for the current loop:** The current loop plant has a single dominant pole (the inductor roll-off). The uncompensated phase at crossover is typically -70° to -80° . A single zero from Type-II is sufficient for 45° PM. Type-III is not realisable on the FAN9672 IEA pin (insufficient component count).

Step 16.3 — Voltage Loop Compensator Design

What needs to be added

- **Type-II vs. Type-III decision:** Show the decision criterion explicitly: If the uncompensated phase at f_{cv} (phase of $T_{v,base}$) is above -90° , Type-II provides sufficient boost. If it is below -90° (which occurs when f_{cv} approaches the LC resonance), Type-III is required. The tool auto-selects based on the computed phase deficit. For this design, [TYPE] was selected because [REASON from calculation].
- **SLVA662 Method B derivation summary:** Method B folds the feedback divider into the compensator transfer function. The OTA network is sized to provide exactly the required gain $1/|T_{v,base}(f_{cv})|$ at the

crossover frequency, with zeros placed below crossover for phase boost and poles above for HF rejection. Show the three-step derivation: (1) compute required $|H_{OTA}|$, (2) compute R_2 from gain requirement, (3) compute C values from zero/pole frequency placement.

- **120 Hz rejection requirement:** The voltage loop must attenuate the 120 Hz component of the bus voltage ripple before it modulates the current reference. Insufficient rejection causes visible harmonic distortion in the input current. 20 dB at 120 Hz is the design floor — explain where this number comes from.

Step 16.4 — Stability Scorecard Interpretation

What needs to be added

- **Why 9 operating points matter:** A design that is stable at the nominal operating point (90 Vac, full load) can be unstable at light load high line (264 Vac, 20% load) due to the drastically different plant gain. The scorecard checks all corners because the real product must be stable across its entire operating envelope.
- **The 45°/60° PM targets — justification:** Current loop $PM \geq 45^\circ$ corresponds to a damping ratio $\zeta \geq 0.38$ — adequate for a current-controlled inner loop. Voltage loop $PM \geq 60^\circ$ is required because the outer loop must reject 120 Hz disturbances — lower PM means the 120 Hz rejection deteriorates.

9. Project & Developer Documentation

Three categories of developer documentation need to exist or be substantially updated. These documents are read by engineers adding new steps, debugging the calculation engine, or deploying the tool to a new environment. Currently they are incomplete or missing entirely.

9.1 — README.md (Root Level)

The root README is the first file any new developer reads. It must answer five questions within two minutes of reading: What is this? How do I run it? How do the pieces fit together? How do I add a step? Where do I find the specifications?

Required sections:

- **Project Overview (2 paragraphs):** What the tool does, what problem it solves, who uses it, what the output is (a 16-step engineering report + interactive GUI).
- **Technology stack table:** Frontend (React 18, TypeScript, Vite), Backend (FastAPI, Python 3.11, ReportLab, NumPy, Matplotlib), Embedded tools (FAN9672 Control Design Tool v4, PFC Inductor Studio V5.1).

- **Architecture diagram (ASCII):**

```

Designer → React GUI (Vite / TypeScript)
↓ HTTP/REST
FastAPI Backend (Python)
■ ■ ■ ■ Mode A: Topology selection engine
■ ■ ■ ■ Mode B: 16-step calculation engine
■ ■ ■ ■ step7_magnetic_calc.py
■ ■ ■ ■ step15_capacitor.py
■ ■ ■ ■ step16_control_design.py
■ ■ ■ ■ generate_combined_report.py → PDF
■ ■ ■ ■ Magnetics DB (JSON)
Embedded JS Tools (srcDoc / public/ iframe)
■ ■ ■ ■ Review Magnetics (review_magnetics.html)
■ ■ ■ ■ Control Design Tool (control_design.html)

```

- **Quick-start (3 commands):** Install, start backend, start frontend. Including the python-multipart dependency note.
- **16-step flow summary table:** One row per step, showing step number, description, calculation file, report section, and frontend component.
- **Environment variables:** VITE_API_URL and any backend configuration.

9.2 — DEVELOPMENT_GUIDE.md (Updated from v2.0)

The current guide is from v2.0 (pre-Steps 15 and 16) and does not describe the patterns established in sessions 1–33. It needs a full update covering:

Required additions:

- **How to add a new step — complete checklist:**

- Create backend/app/mode_b/step{N}_xxx.py (calculation engine)
- Create backend/app/mode_b/generate_step{N}.py (PDF section generator)
- Register the step in generate_combined_report.py
- Add endpoint to main.py
- Add API call to frontend/src/api/client.ts
- Create frontend/src/components/Step{N}Wizard.tsx
- Add step to App.tsx type definition and routing
- Update Stepper.tsx for breadcrumb navigation

- **The approved-result pattern (CRITICAL):** When a step's Approve button stores a result in App.tsx state, that result **MUST** carry a complete selected_xxx block with all part/config data the backend needs to regenerate that step's full report. The Step 15 CapacitorResult.selected_cap field is the reference pattern. Failure to follow this pattern causes silent report truncation.

- **JS tool integration pattern:** The postMessage bridge pattern for embedding JavaScript studio tools (review_magnetics.html, control_design.html). Explains the two-copy sync rule (src/assets/ is the edit copy; public/ is the live served copy), the getDesignState/designState round-trip for extracting JS-computed values, and the srcDoc vs. src= decision criteria.

- **Report generation data flow:** Explains how step15GenerateReport is the master endpoint, how step15_result and step16_params chain together, and how generate_combined_report merges all PDF sections.

- **Testing checklist for each new step:** Unit test for calculation engine, report generation smoke test, frontend navigation test, data flow verification (approved result carries all data for future steps).

9.3 — API Reference

No API reference exists. Every endpoint is currently discoverable only by reading the FastAPI source or the Swagger auto-docs. A human-readable Markdown reference is needed for engineers who need to test or extend the backend.

Required content:

- **Endpoint table:** Method, path, request schema, response schema, which report step it feeds, which frontend component calls it.
- **DesignResult schema:** Full field-by-field documentation of the approved inductor design result object — the most complex data structure in the system.
- **CapacitorResult schema:** Including the selected_cap sub-object.
- **Error convention:** HTTP 500 with string body — how to read backend stack traces.
- **Authentication:** Currently none — document that the API is localhost-only and has no auth layer.

10. Proposed Work Sequence & Effort Estimates

The following sequence is recommended based on: (1) impact on the most-used sections, (2) dependency order (Steps 1–6 establish the tone and template for the rest), (3) risk (higher-risk sections done first while context is fresh).

Priority	Work Item	Files Affected	Est. Effort	Key New Content
1	Steps 1–6 narrative	generate_report.py	Medium	IEC/SEMI context, derivations, 9-point grid rationale
2	Step 16 background + compensator theory	generate_step16.py	High	ACM overview, Bode plot primer, RHP zero, SLVA662 derivation
3	Steps 7–8 background + iGSE + thermal	generate_steps13_14.py	High	Core type narrative, iGSE derivation, SA model, waveform guide
4	Step 15 hold-up + lifetime methodology	generate_step15.py	Medium	Hold-up derivation, ripple current, Arrhenius model
5	Steps 9–12 copper loss + thermal	generate_report.py	Medium	Skin/proximity effect, Litz wire, temperature coefficient
6	README.md	README.md (new)	Low	Project overview, architecture diagram, quick-start
7	DEVELOPMENT_GUIDE.md update	DEVELOPMENT_GUIDE.md	Medium	Add-step checklist, approved-result pattern, JS bridge pattern
8	API Reference	docs/API_REFERENCE.md (new)	Low	Endpoint table, DesignResult schema, CapacitorResult schema

Recommended approach

Work one item at a time in the order above. For each report section, the approach is: (1) draft the new narrative text in plain English, (2) review and approve together, (3) implement in the Python report generator. This prevents rework caused by writing detailed code for content that then needs to change.

11. Open Questions & Design Decisions

The following questions were identified during the planning discussion and need answers before implementation begins. The decision made for each will affect the writing style, depth, and citation format across all sections.

#	Question	Options	Implication
1	External references — formal or informal? Should the report cite IEC 61000-3-2, SEMI F47, Texas Instruments SLVA662, Erickson & Maksimovic, etc. with formal footnote citations?	(a) Formal footnotes — IEEE/IEC style (b) Inline references — "(per SLVA662)" (c) No citations — internal tool only	Option (a) makes the report suitable for customer submission. Option (c) is fastest to implement. Most engineering tools use (b).
2	Starting knowledge level for background sections? Should the college-student-level background assume knowledge of Ohm's Law and basic AC circuits only, or should it also assume knowledge of switching converters?	(a) Start from Ohm's Law (b) Assume basic converter knowledge (c) Assume PFC-specific knowledge	Option (a) adds ~2 pages of primer content per major section. Option (b) is the most practical for an engineering tool. Option (c) is the current state.
3	Target page count for the full report? The current report is approximately 45–55 pages. With full documentation improvements, it will grow. Is there a target or maximum?	(a) No limit — make it complete (b) Target 80–100 pages (c) Keep under 60 pages — add optional appendix	No limit is cleanest architecturally. A page cap forces trade-offs on which content to include. Most comparable design reports for production hardware are 60–120 pages.
4	Diagrams and figures? The current report uses only matplotlib plots for waveforms/Bode. Should block diagrams, schematic excerpts, and conceptual figures be added?	(a) Add matplotlib-generated diagrams (b) Embed static PNG images from specs/ (c) Text-only with ASCII art	Option (a) keeps everything dynamic and auto-generated. Block diagrams (control loop, circuit topology) are best as static images from a drawing tool. Option (b) for non-data figures.
5	Writing style for background sections? Textbook style (third person, passive voice) or engineering report style (first person plural, active voice)?	(a) "The duty cycle is defined as..." (b) "We define the duty cycle as..." (c) "This tool computes duty cycle as..."	Option (a) is the most formal and is standard in published technical reports. Option (c) matches the existing report style and makes it clear the content is auto-generated.

Document Status: Draft — For Review and Discussion. No implementation has begun. This document represents the planning discussion of June 2026 and is subject to revision based on answers to the open

questions above.

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